# Automated Detection of Dark Patterns Using In-Context Learning Capabilities of GPT-3

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***Abstract*—Dark patterns manipulate user choices through deceptive UI tactics. Any automated detection technique for dark patterns must address the varying nature of dark patterns. Existing detection techniques need manual intervention in some cases. In other cases, techniques are not generalized due to overfitting problems; for example, these techniques can not handle cases where texts are semantically similar but possess lexical differences. We propose an automated dark pattern text detection technique that is generalized. We synthesize inclusive definitions of dark pattern categories. This contextual information is prioritized using in-context learning capabilities of GPT-3 to detect and classify dark pattern texts. Results show that our technique offers satisfactory performance for 6 out of 7 dark pattern categories explored in this study. We also validate the improved generalization capability of our technique by outperforming an existing baseline model on a test dataset.**

***Index Terms*—Dark Patterns, Automated Detection, Large Language Models, GPT-3, In-Context Learning**

## I. Introduction

Dark patterns are deceptive user interface (UI) design tactics that abuse knowledge of psychology to alter user decision-making [1]. For example, online services often make it difficult to unsubscribe from their service (‘Obstruction’ dark pattern), which leads many users to change their decision. This kind of tactics can harm users in the form of financial loss, sharing of personal data, and induction of compulsion and addiction [2]. As these tactics are deceptive in nature, often users can not identify what is wrong, even when they feel negative emotions because of it.

Automated detection of dark patterns can help users identify and beware of such deceptive design tactics, while also facilitating large-scale collection of dark pattern related data. However, detecting dark patterns automatically is challenging because of their varying representations in UI. Detection techniques must have some degree of generalization to be useful to real-life users. That is why we need a mechanism that can process contextual information in a human-like manner. Only then it will be possible to cope with the varying nature of dark patterns.

There are different approaches to automated dark pattern detection. Images or screenshots of UI can be processed to detect dark patterns. However, such a technique only detects the existence of a correlation between pixels, textual information is ignored [3]. Another way is to process the UI texts using machine learning (ML) or rule-based mechanisms. However, rule-based mechanisms are difficult to generalize as they only handle specific cases. Yada et al. [4] proposed a technique to detect dark pattern texts by training machine learning models with a labelled textual dataset. However, such an approach may be vulnerable to overfitting if the dataset used for training can not be generalized. Despite recent advancements, no automated detection approach processes contextual information in a human-like manner.

We propose a novel detection technique for dark pattern texts that can be generalized. To improve generalization, We synthesize inclusive definitions of dark pattern categories as contextual information. This contextual information is prioritized, along with zero, one, or two examples per category, to engineer prompts for large language models (LLM). In-context learning capabilities of LLMs like GPT-3 enable them to perform a new task using inference alone by conditioning on a few examples [5]. We use our engineered prompts to leverage this capability in detecting dark pattern texts.

Our approach is validated by testing on the combined dark pattern text dataset of Mathur et al. [6] and Yada et al. [4]. In terms of detection accuracy, we achieve satisfactory results for 6 out of 7 dark pattern categories and also the non dark pattern category. Using at most only two examples per class makes our approach less prone to overfitting. To further validate that our approach can be generalized on other datasets, we compare our results with the results of Yada et al. [4] on another dataset manually labelled by us. Our approach outperforms the results of Yada et al. [4] on this test dataset.

## II. Related Work

Dark patterns have received attention from the research community very recently, which is growing day by day. Researchers have identified dark patterns and their taxonomy [1], studied user perception of dark patterns [7], and explored policy and regulatory perspectives [8]. At the same time, they studied techniques for detecting dark patterns [9] [10], both manually and automatically.

Defining and classifying dark patterns has been challenging because of the varying ways such deceptive techniques are manifested in UIs. Brignull [11] coined the term ‘dark patterns’ and defined a set of “tricks used in websites that

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make a user do things that the user did not mean to". Later, Gray et al. [1] proposed a taxonomy of five such design tricks based on user experience. Mathur et al. [6] also proposed a similar taxonomy of seven broad categories of dark patterns in their work. In addition, researchers have also defined and classified dark patterns based on privacy [12], cognitive biases that dark patterns use [6], attention-capture dark patterns [13], dark patterns found on video streaming platforms [14], and proxemic (i.e., physical proximity) interaction [15]. However, these works are often distinct, making it a complex undertaking to unify the existing body of literature.

Without a holistic definition and classification of dark patterns, detecting such design techniques becomes quite a complex task. Mathur et al. [6] used a semi-automatic technique to identify and show the existence of dark patterns in shopping websites. They analyzed 11,000 websites to show the prevalence of dark patterns. Some parts of their detection technique were automated, but the final work of detecting dark patterns from texts was done manually. Yada et al. [4] provided a dataset of dark pattern texts extracted from e-commerce websites that could be used in machine learning (ML) classifiers to detect dark patterns. They also applied some classical NLP methods (SVM, random forest etc.) and transformer-based pre-trained language models (BERT, RoBERTa etc.) to show the accuracy of automated detection on the dataset. However, their models may suffer from overfitting issues, resulting in reduced generalization. Moreover, they used binary classification, i.e., their approach did not classify dark pattern texts into specific categories.

Soe et al. [3] used supervised ML to train a prediction model to identify dark patterns in cookie banners. However, this is also a semi-automated technique because a user has to manually input the characteristics of a UI for the ML model to generate a prediction. Mansur et al. [9] proposed an automated approach that uses computer vision and natural language processing techniques to detect dark patterns. However, their text analysis technique is solely based on heuristically defined pattern-matching rules, making its application on semantically complex texts difficult. Thus, the existing literature shows that automated detection of dark patterns is a challenging task, often requiring some degree of manual intervention or suffering from a lack of generalization.

As this is still a new field of research and detection tools are yet to mature, large-scale collections of dark pattern related data are rarely available. Moreover, existing works vary greatly in data representation. Some researchers analyze dark patterns with screenshots of UIs [9], whereas others have used feature-based techniques [3]. Mathur et al. [6] collected the first large-scale textual dark pattern data. 1,818 dark pattern texts were found by analyzing 11,000 shopping websites. Yada et al. [4] extended this dataset by adding 1,178 non-dark pattern texts from the same websites.

## III. Methodology

We use a novel approach to solve the challenging problem of automated dark pattern detection that can be generalized. We start with synthesizing inclusive definitions of dark pattern categories. Then we prioritize this contextual information to engineer prompts for large language models (LLM). LLMs like GPT-3 have in-context learning capabilities [5] that we leverage in detecting dark pattern texts with a reduced risk of overfitting.

### A. Dataset

We use the combined dataset of Mathur et al. [6] and Yada et al. [4] that contains 1,178 dark pattern and 1,178 non-dark pattern texts, totaling 2,356 texts from e-commerce websites. Dark pattern texts in this dataset are labeled with 7 broad categories of dark patterns - 'Misdirection', 'Urgency', 'Scarcity', 'Social Proof', 'Obstruction', 'Forced Action', and 'Sneaking'. The non-dark pattern texts are labeled with the 'Not Dark Pattern' category. Definitions of these 8 categories are formulated for in-context learning, which is elaborated in the next section. An example of 'Urgency' from the online food ordering platform Foodpanda [16] is shown in Fig. 1. Of the 2,356 instances of the dataset, 2,342 are exclusively used to validate our classification models. The other 14 texts are used for learning or validation based on prompting techniques used for GPT-3.

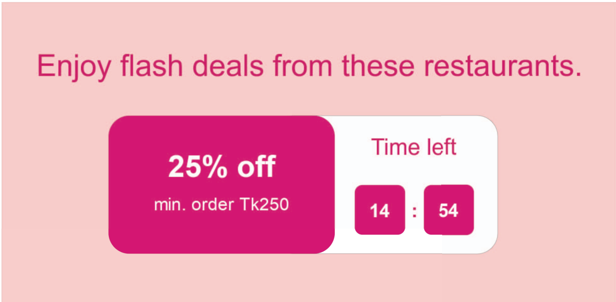


Fig. 1. Example of Urgency on Foodpanda [16]

### B. Defining Dark Pattern Categories

We must first define the dark pattern categories to classify texts into those categories. As we want to prioritize this contextual information in our classification task, the meaning and wording of the category definitions can greatly impact the results. Each dark pattern category may have multiple variants. Instead of using the high-level definitions of categories, we create the definitions for each category using the definitions of the corresponding variants. For this purpose, we synthesize relevant information from literature [1] [6] [9] [11] to create inclusive definitions for the corresponding variants of each category. Then we enumerate the single-quoted definitions of the variants to create definitions for the categories. The resulting definitions are listed in TABLE I. Our definition of the term 'dark patterns' is also synthesized from the same sources. We define 'dark patterns' as "user interface design techniques that use knowledge of human behavior to benefit the service provider by coercing, steering, tricking or deceiving

TABLE I
DEFINITIONS OF CLASSIFICATION CATEGORIES

| **Classification Category** | **Definition** |
|---|---|
| Misdirection | 'Misdirection' includes 'attracting the user's attention to one thing in order to distract their attention from another' and 'use of visuals, confusing language, and emotion (shame) to intrigue users to or away from particular choices'. |
| Urgency | 'Urgency' means 'pressuring the user to accept deals or discounts with time limitations, count-down timers, or sale deadlines'. |
| Scarcity | 'Scarcity' means 'pressuring the user to accept deals or discounts with indications of low stock, limited supply, high demand, or low availability'. |
| Social Proof | 'Social Proof' includes 'informing the user about the activity on the website (e.g., purchases, views, visits) to make the product more credible' and 'testimonials on a product page whose origin is unclear'. |
| Obstruction | 'Obstruction' includes 'making a process more difficult than it needs to be, with the intent of obstructing certain actions' and 'making it easy for the user to sign up or subscribe for a service but hard to cancel it'. |
| Forced Action | 'Forced Action' means 'requiring the user to perform an undesirable action (e.g., creating accounts or sharing information) to access certain functionality'. |
| Sneaking | 'Sneaking' includes 'hiding or delaying the relevant information from the user', 'adding additional products to shopping carts without their consent', 'revealing previously undisclosed charges to users right before the checkout' and 'enrolling users in a recurring subscription or payment plan without clear disclosure or their explicit consent'. |
| Not Dark Pattern | If a text does not fall into the other 7 categories, it is defined as 'Not Dark Pattern'. |

users into making decisions that, if fully informed and capable of selecting alternatives, users might not make" [1] [6] [11].

### C. Classification Using GPT-3

We use the 'GPT for Sheets™ and Docs™' extension available in 'Google Sheets' to classify all the texts in our dataset. This extension provides useful GPT functions to be used in a spreadsheet. We use the 'GPT()' function that submits a prompt to GPT and returns the answer. This function has a mandatory parameter 'prompt', and three optional parameters - 'value', 'temperature' and 'model'. These parameters can be tuned to get the desired results.

The parameter 'temperature' indicates whether the language model would prefer creativity or precision in its answers. It takes a value between 0 and 1, where lower values mean it would prefer precision and higher values mean it would use creative freedom in its answers. We set the temperature to 0 because classification into dark pattern categories requires precise answers. The parameter 'model' indicates the specific language model we want to use. We use the 'gpt-3.5-turbo' model in our study. We use the mandatory parameter 'prompt' to provide GPT with the contextual information. The optional parameter 'value' indicates the value to be concatenated at the end of the prompt. To classify one single UI text, we provide that text in this 'value' parameter. In this way, we call the 'GPT()' function for all the UI texts in our dataset.

### D. Prompt Engineering

We engineer our prompts to tune the large language model of GPT-3 for dark pattern classification. Our prompts have three parts - 'Context', 'Input', and 'Output'. 'Input' lets the model know what to expect as input and what to do with it, while 'Output' defines the answer format. We engineer the 'Context' to apply in-context learning for the following techniques - zero-shot, one-shot, and few-shot prompting.

*1) Zero-Shot Prompting:* In zero-shot prompting, no example for any of the classification categories is used. A template for zero-shot learning is given below -

**Prompt:**
Context
<Definition of Dark Patterns>
<Classification of Dark Patterns>
<Definition of Category 1>
..............................................
..............................................
<Definition of Category 7>
<Definition of Category 8 (Not Dark Pattern)>
Input
<Input Format><Processing Instruction>
Output
<Output Format><Text to be classified>
**GPT-3 Response:**
<Predicted Category>

*2) One-Shot Prompting:* In one-shot prompting, exactly one example is provided for each of the 7 dark pattern categories, along with the definitions. As the 'Not Dark Pattern' category includes everything that does not fall into the other 7 categories, providing one specific example for this category might bias the language model towards that particular example. Thus, we do not provide an example for the 'Not Dark Pattern' category. Each <Definition of Category X> is followed by <Example 1 of Category X>, except for the last category (Not Dark Pattern). The first example for each category shown in TABLE II is used in one-shot prompting. The 7 example texts are randomly selected and removed from the validation dataset.

*3) Few-Shot Prompting:* Few-shot prompting is similar to one-shot, except more than one example must be provided. We provide two examples to GPT-3, along with the definitions.

TABLE II
EXAMPLE TEXTS USED FOR EACH DARK PATTERN CATEGORY

| Dark Pattern Category | Examples |
|---|---|
| Misdirection | 1. No, I'll rather pay full price<br>2. No thanks, I dont want a discount |
| Urgency | 1. Items reserved for 15:00<br>2. 1 day 08:15:25 |
| Scarcity | 1. 1 LEFT<br>2. 87% offers claimed. Hurry up! |
| Social Proof | 1. IN 43 PEOPLE'S SHOPPING BAG<br>2. Armin Dinovic bought 10M Runescape 3 Gold Order total: 3,70€ About 10 seconds ago |
| Obstruction | 1. You may change the items in your order, or cancel the Smartship at anytime, up until 3 days prior to the scheduled ship date of your Smartship by calling Customer Service at 1-800-518-0284<br>2. We may also disclose your information to third parties who may contact you with details of other products and services which may be of interest. If you do not want your name and mailing details made available in this way please email opt-out@nextdirect.com |
| Forced Action | 1. I would like to join Backstage Pass & agree to the Terms & Conditions & to receive emails & other promotional offers<br>2. I agree to receive marketing emails from Natural Life and agree to our Privacy Policy and terms of use |
| Sneaking | 1. Order Subtotal $19.99 Standard Delivery $12.99 Care & Handling $2.99 Tax $2.38 Total $38.35 Savings Today $10.00<br>2. Purchase protection added |

Like in the one-shot learning case, example for the 'Not Dark Pattern' category is not provided. Each <Definition of Category X> is followed by <Example 1 of Category X> and <Example 2 of Category X>, except for the last category (Not Dark Pattern). The first 7 example texts used for 7 dark pattern categories are the same as one-shot, while the other 7 example texts are randomly selected. All of these 14 texts are shown in TABLE II. They are removed from the validation dataset.

## IV. EVALUATION

Accuracy, precision, recall and F1 score are calculated category-wise to assess the performance of our dark pattern classification. However, accuracy and recall give a better understanding of how our technique performed in classifying texts from a particular category because the precision (and thus, F1 score) of a category is a concern for the performance of other categories (impacting their accuracy and recall), not the category in concern. These results are listed in TABLE III. We also validate how generalized our detection technique is, using an additional test dataset of dark pattern texts.

### A. Result Analysis

We subject the entire validation dataset of 2,356 texts to classification using zero-shot prompting. The model accurately classifies 1,969 texts, while 387 texts are misclassified. This translates to an overall accuracy of 83.57%. The model performs significantly better in classifying texts from categories such as 'Forced Action', 'Urgency', 'Scarcity', 'Social Proof', and 'Not Dark Pattern'. In contrast, it performs very poorly when dealing with categories like 'Misdirection', 'Sneaking', and 'Obstruction', indicating the limitations of relying solely on definitions for classification, without any examples.

2,349 texts are classified using one-shot prompting. The model accurately classifies 2,137 texts, while 212 texts are misclassified. This translates to an overall accuracy of 90.97%. Like the zero-shot model, the one-shot model also performs significantly better in classifying texts from the same 5 categories. Even though the model performs poorly in the 'Sneaking' category, it performs significantly better than the zero-shot model in classifying texts into 'Misdirection' and 'Obstruction' categories. Accuracy of the 'Misdirection' category rises from 6.67% in zero-shot to 53.61% in one-shot, and the 'Obstruction' category rises from 11.11% in zero-shot to 84.62% in one-shot. This insight indicates that providing more examples for these categories might improve performance. One possible reason behind this is the wide variation in semantics for the instances of these categories.

Few-shot prompting is tested on 2,342 texts. The model accurately classifies 2,168 texts, while 174 texts are misclassified. This translates to an overall accuracy of 92.57%. Like the other two models, this model also can not classify texts from the 'Sneaking' category. This category mostly leverages website logic and policy, rather than individual webpage texts. As a result, this category of dark patterns is challenging to detect only with textual information. The definition of this category provided in TABLE I also conforms to our findings. Few-shot prompting performs well for all the other 7 categories of texts. It classifies texts from the 'Obstruction' category with an accuracy of 92%, which is a huge increase from 11.11% in zero-shot and is higher than 84.62% in one-

TABLE III
EXPERIMENTAL RESULTS USING GPT-3

| | Zero-Shot | | | | One-Shot | | | | Few-Shot | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Category** | **Accuracy** | **Precision** | **Recall** | **F1 Score** | **Accuracy** | **Precision** | **Recall** | **F1 Score** | **Accuracy** | **Precision** | **Recall** | **F1 Score** |
| Misdirection | 6.67% | 44.83% | 6.67% | 11.61% | 53.61% | 97.2% | 53.61% | 69.1% | 65.8% | 97.69% | 65.8% | 78.64% |
| Urgency | 93.33% | 73.13% | 93.33% | 82.01% | 92.34% | 84.65% | 92.34% | 88.33% | 93.27% | 90.65% | 93.27% | 91.94% |
| Scarcity | 92.34% | 87.73% | 92.34% | 89.98% | 95.92% | 93.68% | 95.92% | 94.79% | 98.56% | 92.34% | 98.56% | 95.35% |
| Social Proof | 92.63% | 98.3% | 92.63% | 95.38% | 96.46% | 98.68% | 96.46% | 97.56% | 94.84% | 99.32% | 94.84% | 97.03% |
| Obstruction | 11.11% | 75% | 11.11% | 19.35% | 84.62% | 78.57% | 84.62% | 81.48% | 92% | 82.14% | 92% | 86.79% |
| Forced Action | 100% | 6.56% | 100% | 12.31% | 100% | 6.12% | 100% | 11.54% | 100% | 4.35% | 100% | 8.33% |
| Sneaking | 0% | 0% | 0% | 0% | 0% | 0% | 0% | 0% | 0% | 0% | 0% | 0% |
| Not Dark Pattern | 91.51% | 85.62% | 91.51% | 88.47% | 94.65% | 92.45% | 94.65% | 93.54% | 94.91% | 94.59% | 94.91% | 94.75% |

shot. It further increases the results of the one-shot model for the 'Misdirection' category from 53.61% to 65.8%.

GPT-3 performs well in 5 of the 8 categories for all three prompting techniques. Providing more examples in cases of 'Misdirection' and 'Obstruction' categories improves the classification by significant margins.

### B. Generalization

We use at most only two examples per class to achieve our results. This is already enough evidence that our technique will be less prone to overfitting and can be generalized. We go one step further to validate this. We compare our dark pattern detection performance with that of Yada et al. [4] on a test dataset prepared by ourselves. As they used an e-commerce dataset in their work, we extract dark pattern texts from 30 Bangladeshi e-commerce websites and manually label them to create the test dataset. On this new test dataset, we compare our results with that of the SVM model provided by them. It detected 92.2% of dark pattern texts in their own dataset which decreases to 42.8% in the new test dataset. Our model using few-shot prompting performs significantly better, detecting 58.67% of dark pattern texts from the test dataset.

## V. THREATS TO VALIDITY

Evaluation on only two datasets poses an external validity threat as results may vary based on test data sources. However, contextual information is prioritized in prompt engineering to mitigate this threat. Random selection of example texts may affect the internal validity. There is scope to incorporate other sampling techniques that may change the results.

## VI. CONCLUSION AND FUTURE WORK

Our automated approach for detection of dark pattern texts uses in-context learning capabilities of GPT-3 to tackle varying nature of dark patterns. Results show that our technique detects 6 out of the 7 dark pattern categories successfully and can be generalized to other sources of dark pattern data as well. Using in-context learning allows our detection technique to reduce the risk of overfitting and gives emphasis on the contextual information regarding dark patterns. As a result, this technique can be as much generalized as the contextual information provided during learning. Our future work includes improving the performance of this technique by creating a more detailed and specific definitions of dark pattern categories.

## REFERENCES


[1] C. M. Gray, Y. Kou, B. Battles, J. Hoggatt, and A. L. Toombs, "The dark (patterns) side of UX design," in *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, apr 2018.

[2] A. Mathur, M. Kshirsagar, and J. Mayer, "What makes a dark pattern... dark?" in *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, may 2021.

[3] T. H. Soe, C. T. Santos, and M. Slavkovik, "Automated detection of dark patterns in cookie banners: how to do it poorly and why it is hard to do it any other way," *arXiv preprint arXiv:2204.11836*, 2022.

[4] Y. Yada, J. Feng, T. Matsumoto, N. Fukushima, F. Kido, and H. Yamana, "Dark patterns in e-commerce: a dataset and its baseline evaluations," in *2022 IEEE International Conference on Big Data (Big Data)*. IEEE, dec 2022.

[5] S. Min, X. Lyu, A. Holtzman, M. Artetxe, M. Lewis, H. Hajishirzi, and L. Zettlemoyer, "Rethinking the role of demonstrations: What makes in-context learning work?" *arXiv preprint arXiv:2202.12837*, 2022.

[6] A. Mathur, G. Acar, M. J. Friedman, E. Lucherini, J. Mayer, M. Chetty, and A. Narayanan, "Dark patterns at scale," *Proceedings of the ACM on Human-Computer Interaction*, vol. 3, no. CSCW, pp. 1–32, nov 2019.

[7] L. D. Geronimo, L. Braz, E. Fregnan, F. Palomba, and A. Bacchelli, "UI dark patterns and where to find them," in *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, apr 2020.

[8] J. Luguri and L. Strahilevitz, "Shining a light on dark patterns," *SSRN Electronic Journal*, 2019.

[9] S. M. H. Mansur, S. Salma, D. Awofisayo, and K. Moran, "AidUI: Toward automated recognition of dark patterns in user interfaces," in *2023 IEEE/ACM 45th International Conference on Software Engineering (ICSE)*. IEEE, may 2023.

[10] I. Stavrakakis, A. Curley, D. O'Sullivan, D. Gordon, and B. Tierney, "A framework of web-based dark patterns that can be detected manually or automatically," 2021.

[11] H. Brignull, M. Leiser, C. Santos, and K. Doshi, "Deceptive patterns – user interfaces designed to trick you," 2023. [Online]. Available: https://www.deceptive.design/

[12] L. Jarovsky, "Dark patterns in personal data collection: Definition, taxonomy and lawfulness," *SSRN Electronic Journal*, 2022.

[13] A. M. Roffarello and L. D. Russis, "Towards understanding the dark patterns that steal our attention," in *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. ACM, apr 2022.

[14] A. Chaudhary, J. Saroha, K. Monteiro, A. G. Forbes, and A. Parnami, ""are you still watching?": Exploring unintended user behaviors and dark patterns on video streaming platforms," in *Designing Interactive Systems Conference*. ACM, jun 2022.

[15] S. Greenberg, S. Boring, J. Vermeulen, and J. Dostal, "Dark patterns in proxemic interactions," in *Proceedings of the 2014 conference on Designing interactive systems*. ACM, jun 2014.

[16] "Foodpanda: Food and more, delivered." [Online]. Available: https://www.foodpanda.com/